

8 Concurrency

There are many angles to understanding concurrency in computer programming. We won't explore them all (that's a class all on its own), but we'll look at concurrency in the context of creating responsive gui-based applications.

An RSS reader features an underlying engine or model that did the work of parsing RSS feeds. The GUI or view merely reflects the data contained in the model. Whenever the model updates, we update the view.

We assume that the process of updating the model was quick so that the view could simply wait for the model to update before doing its own work. However, what happens when the model takes a long amount of time to update?

```
public class SlowForm : Form
{
    Button button;
    TextBox input;
    public SlowForm()
    {
        Size = new Size(200, 200);

        input = new TextBox();
        input.Size = new Size(100, 25);
        input.Location = new Point(35, 50);

        button = new Button();
        button.Size = new Size(100, 25);
        button.Location = new Point(35, 100);
        button.Text = "Do Work";
        button.Click += new EventHandler(delegate(object
sender, EventArgs e)
        {
            // Simulate some gnarly work to do by
a boring loop
            for (int i = 0; i < 1000000000; i++) { }
            MessageBox.Show("Done!");
        });
        this.Controls.Add(button);
    }
}
```